SUMMARY · MK 4 OPERATIONS & SUPPLY CHAIN MANAGEMENT · WEEK 7

Waiting Lines and Simulation

Waiting lines and simulation (Ch. 10) through agent-based and generative agent-based modelling, with the KAI Commuter Manggarai Station case (and references to Gloria Hospital and the Massachusetts storage initiative). Covered: why lines form and the service vs waiting cost trade-off, variability as the engine of queues, five structural choices, the exponential distribution, Jacobs' four models and the two-moment approximation, worked examples, Little's Law, queue explosion at high utilisation, Bob & Ray (blocking and starving), when formulas fail, DES, ABM, GABM, and four Manggarai discussion questions.

Material

1 Learning objectives and roadmap

1. Understand the waiting line problem and why the cost of waiting is hardest.
2. Read a queuing system's structure.
3. Apply the four Jacobs models and the two-moment approximation.
4. Explain when formulas fail and what DES, ABM and GABM add.
5. Apply to live cases.

2 Why lines form and can't simply be removed

- Weigh the cost of faster service against the cost of waiting.
- Lines form even when average capacity exceeds average demand — randomness.
- Removing lines means buying peak capacity that idles most of the day.

- The decision is economic, not technical.

Cost of service
Observable.

Cost of waiting
Often unpriceable.

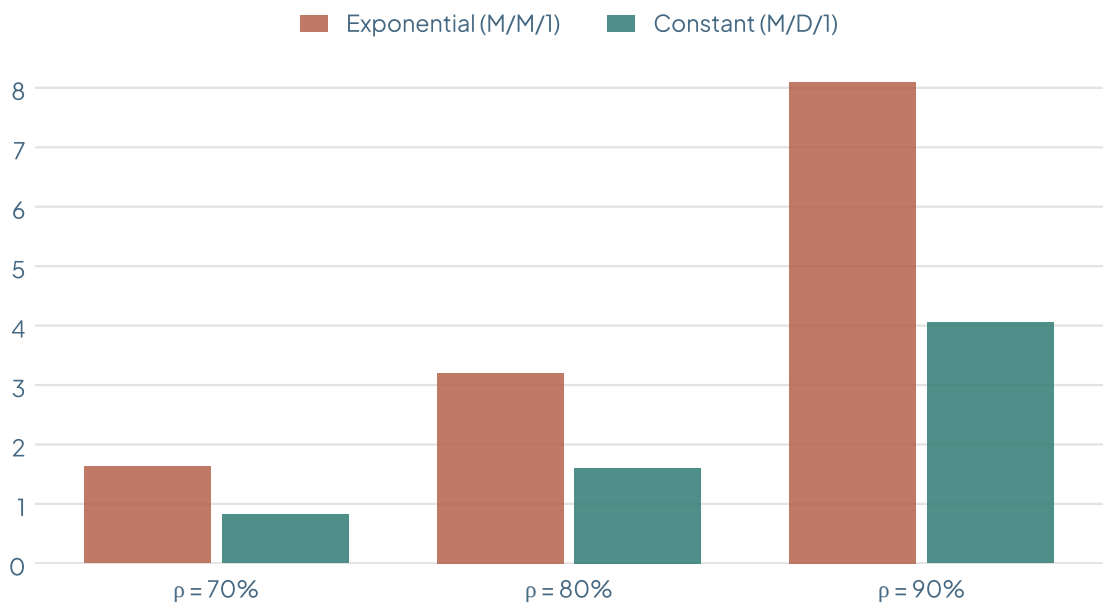
Consequence
Internal waiting is arithmetic; human waiting needs judgement.

3 Variability is the engine of the queue

Factor	Explanation
Arrival variability	Poisson: random, independent, memoryless.
Service variability	Exponential: mostly short, few very long; constant service halves the queue.
Utilisation $\rho = \lambda/\mu$	Queue explodes as $\rho \to 1$.
Joint effect	Lq scales with the sum of squared CVs.

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Two-moment (Kingman): Wq ≈ [ρ ÷ (1 - ρ)] × [(Ca² + Cs²) ÷ 2] × service time
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Lq: exponential vs constant service



Key idea

Exponential vs constant service differ by a factor of two in waiting — a free halving.

4 The anatomy of a queuing system: five structural choices

Choice	Explanation
Source population	Finite or infinite (finite → Model 4)
Arrival pattern	Controllable or not; Poisson default
Queue discipline	FCFS assumed
Service configuration	Channels in parallel, phases in series
Exit	Matters when the population is finite

The exponential distribution is memoryless with a long right tail; it breaks for scheduled arrivals. Map the structure correctly or every later number is meaningless.

5 Jacobs' four waiting line models and their formulas

Model	Structure	Population	Service
1	Single channel	Infinite	Exponential
2	Single channel	Infinite	Constant
3	Multichannel	Infinite	Exponential
4	Single/multi	Finite	Exponential

Model	Formulas
1	$\rho = \lambda/\mu \cdot Lq = \lambda^2/[\mu(\mu - \lambda)] \cdot Ls = \lambda/(\mu - \lambda) \cdot Wq = Lq/\lambda \cdot Pn = (1 - \rho)\rho^n$
2	$Lq = \lambda^2/[2\mu(\mu - \lambda)] \cdot Ls = Lq + \lambda/\mu$
3	Lq from tables $\cdot Ls = Lq + \lambda/\mu \cdot Wq = Lq/\lambda$
4	$X = T/(T + U)$; D, F from finite tables; $L = N(1 - F)$; $H = FNX$; $J = NF(1 - X)$

$$\text{Little's Law: } L = \lambda W$$

Use it

30 patients waiting, 12 arriving per hour → W = 2.5 hours.

6 Example 10.1: Western National Bank drive-thru

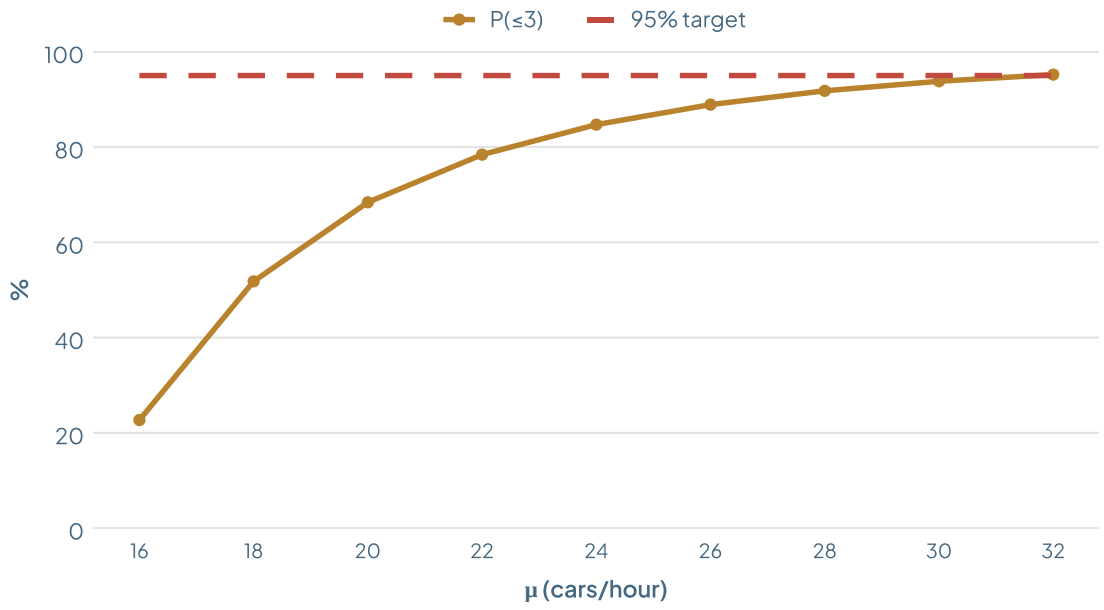
Model 1: $\lambda = 15$/hour, $\mu = 20$/hour.

1. $\rho = 75\%$
2. $Lq = 225 \div 100 = 2.25$ cars
3. $Wq = 0.15$ h = 9 minutes

4 Ls = 3, Ws = 12 minutes

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P(n ≤ 3) = 1 - ρ⁴ = 68.4% at ρ = 0.75; for 95%, ρ ≤ 0.473 → μ ≈ 32
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P(≤3 cars) vs service rate μ (λ = 15/h)



Service capability must rise 60%, and the teller idles 53% of the time.

7 Three more worked decisions

Decision	Model	Result
Robot Car Wash equipment	2	Unit I 12.5-minute wait vs 5-minute tolerance; Unit II 4 minutes
Glenn-Mark servers	3	A 4th clerk costs $240/day, saves $342.20 of mechanics' time
Loose Knit	4	Four machines, one repairer, finite tables
Bank service level	1	95% → μ ≈ 32

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Unit I: λ = 10, μ = 12 constant → Lq = 2.08, Wq = 12.5 min
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8 Why queues explode at high utilisation, and Bob & Ray

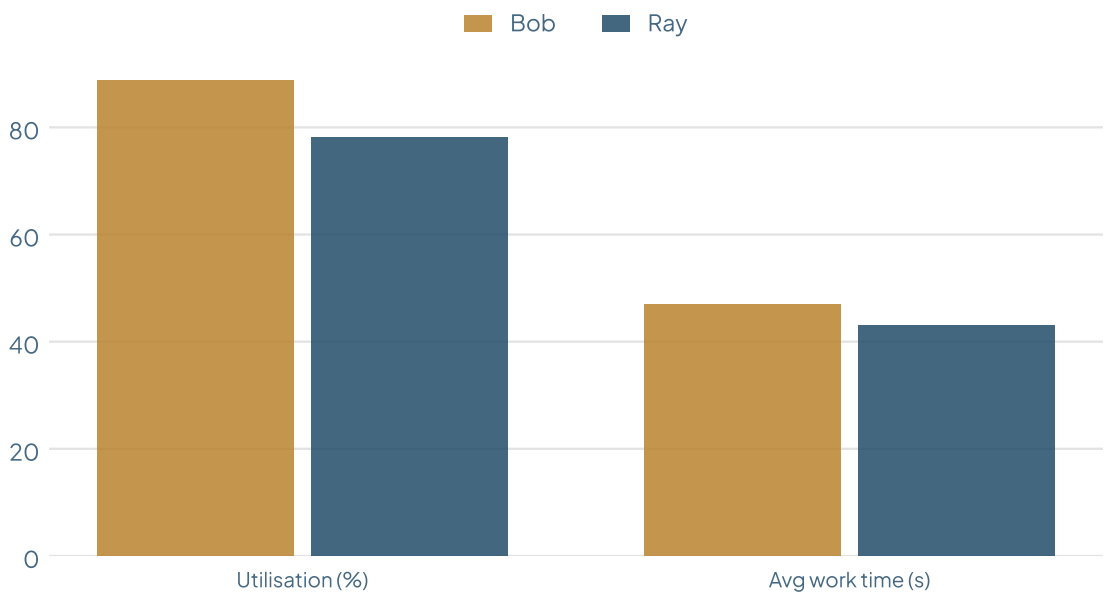
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Lq = ρ² ÷ (1 - ρ): 0.50 at 50%, 1.63 at 70%, 4.82 at 85%, 18.05 at 95%
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Heart of it

85% is not "15% spare" — the queue is already several times larger.

Bob & Ray: two dependent stages cause blocking (Bob waits on Ray) and starving (Ray idles); formulas fail, so simulate. Bob 88.7% utilised (47 s average), Ray 78.2% (43 s), output 60 s per unit.

Bob & Ray simulation (10 units)



Simulation gives the whole distribution, including tails and transient behaviour.

9 The assumptions, and where they break

Formulas are steady-state and break when rates change over time.

Assumed

- Poisson arrivals independent of the queue
- FCFS, constant rates
- Homogeneous, passive customers

vs

Reality

- Balking and reneging
- Rates swing by hour
- Jockeying and abandonment

Formulas → Spreadsheet sim → DES → ABM → GABM → RL agents

Tractability falls as realism rises; validation gets harder; start at the left and move right only when forced.

10 Discrete-event simulation (DES)

Term	Meaning
Entity	What flows
Attribute	Its properties
Event	Changes state
Resource	What entities compete for
State	Current condition
Clock	Jumps event to event

Simulation buys transient behaviour, behavioural rules and complex topology, at the cost of distributions, replication, validation, hidden assumptions and budget. Applications: restaurants/hospitals; a Greek fitness-equipment plant (lead time −1/3, waiting 6.8 → 3.5 days); Alzheimer HTA in Thailand (donepezil). In

chained queues, departures of one stage become arrivals of the next, so Poisson does not survive.

11 Agent-based modelling (ABM) and generative ABM

ABM: agents with rules in an environment, stepping by tick; output is emergent patterns. Schelling (1971): a mild local rule produced full segregation — the information lived in the interactions.

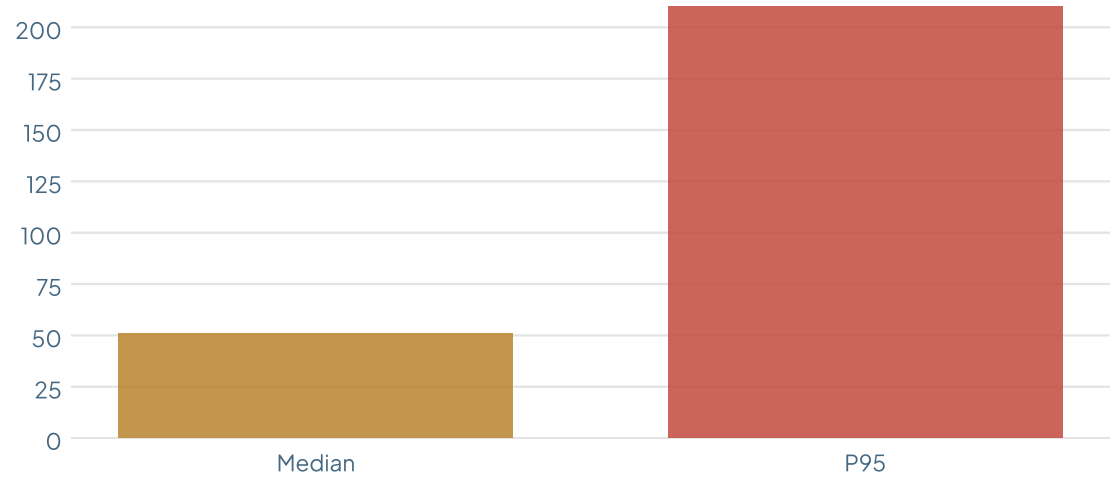
Aspect	DES	ABM
Who decides	Passive entity	Deciding agent
Question	How long the wait given a configuration	What configuration emerges
Output	Utilisation, throughput, wait	A pattern

GABM: profile/state, memory, reasoning (LLM prompt), interaction interface, generated action. Strengths: population synthesis, epidemiology, social phenomena, digital economy. Cautions: no closed-form validation, stochastic and version-dependent, costly, may reproduce stereotypes. For Tuesday staffing, Model 3 still wins.

12 The Manggarai case: the queue at the largest junction

Indonesia's busiest transit junction: 162,000 movements/day, 57.67m transit passengers in 2024, 739 services/day, corridor utilisation 0.84 at peak. Four queues in series: gates (Model 2), ticket counters (Model 3), the vertical transfer corridor (flow capacity), the platform (batch service — none of the four models).

Manggarai: median vs P95 corridor wait (s)



Ratio 4.1 vs ~5.4 for M/M/1 at ρ 0.84; arrival waves prove the variability.

Discussion: (1) Median 51 s vs P95 210 s — M/M/1 itself gives a P95/median of ~4.3 for those who wait, so the ratio alone doesn't prove excess variability; wave arrivals after each train and an empty-60%/blocked-15% corridor do, as does the day-to-day comparison. (2) Most of the five 5 Oct 2024 interventions act on λ, because μ and S lie outside station control. (3) Answer scheduling now with arithmetic on eight weeks of data; build DES for the capital question. (4) Set the unpriced service standard as a constraint before costing.

13 Case close: five takeaways

1. Averages hide the queue.
2. Evidence must match the claim.
3. Constraints set the instruments.
4. Escalate complexity only when forced.
5. The unpriced needs protecting — set the service standard as a constraint first.

Key takeaways

1. Lines form from randomness; the decision is service cost vs waiting cost.
2. Variability drives queues; constant service halves them; Kingman approximation.
3. Five structural choices.
4. The exponential distribution is memoryless.
5. Four Jacobs models; simulate beyond them.
6. M/M/1 and M/D/1 formulas; Little's Law.
7. Western National Bank example.
8. Queues explode at high utilisation.
9. Bob & Ray: blocking and starving.
10. Assumptions and where they break.
11. DES vocabulary and value.
12. ABM and GABM.
13. Manggarai lessons.

Case studies

Case & worked answers

KAI Commuter & the Manggarai capacity dilemma

Four queues in series: gates (Model 2), counters (Model 3), the corridor (flow capacity) and platforms (batch service). Q1: an M/M/1 at ρ=0.84 gives a P95/median ratio ≈ 5.4, so the observed 4.1 alone doesn't prove excess variability; wave arrivals and a corridor empty 60% / jammed 15% do — a variability problem. Q2: four of five interventions act on λ (security, partitioning, stop-and-go, extra trains spreading load); portable counters add S — actions map constraints. Q3: scheduling questions can use existing data; capital

questions need DES. Q4: set a service/safety standard as a constraint before costing options.

Worked examples

Problem 1 · Medium **Antrean M/M/1**

Kasir: rata-rata 24 pelanggan datang per jam (λ), kasir melayani 30 per jam (μ). Hitung utilisasi, panjang antrean, waktu tunggu, dan waktu total di sistem.

1. $\rho = \lambda \div \mu = 24 \div 30 = 0{,}8$ (80%).
2. $$L_q = \lambda^2 \div [\mu(\mu - \lambda)] = 576 \div (30 \times 6) = 3{,}2 \text{ orang}$$
3. $W_q = L_q \div \lambda = 3{,}2 \div 24 = 0{,}133$ jam = 8 menit.
4. $W_s = 1 \div (\mu - \lambda) = 1 \div 6$ jam = 10 menit (8 menit antre + 2 menit dilayani).
5. Peluang kasir kosong $P_0 = 1 - \rho = 20\%$.

Answer: ρ 80%, Lq 3,2 orang, Wq 8 menit, Ws 10 menit.

Problem 2 Easy **Little's Law**

Sebuah stasiun dilewati 120 penumpang per jam; tiap penumpang rata-rata berada di stasiun 15 menit. Berapa rata-rata jumlah orang di stasiun?

1. $L = \lambda \times W$
2. Samakan satuan: W = 15 menit = 0,25 jam.
3. L = 120 × 0,25 = 30 orang.

Answer: Rata-rata 30 orang di stasiun.

Practice questions

Try answering first. The answer key is below each question.

1. λ 15, μ 20. Lq = …

A. 0.75
B. 2.25
C. 3
D. 9

Answer: B. 225/100.

2. Little's Law states…

A. $L = \lambda W$
B. $L = \mu - \lambda$
C. $W = \lambda/\mu$
D. $\rho = \mu/\lambda$

Answer: A. Any stable system.

3. At equal utilisation, constant service gives a queue…

A. Same
B. Double
C. Half
D. Zero

Answer: C. Half.

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4. With a small source population (4 machines)…

A. Model 1
B. Model 2
C. Model 3
D. Model 4

Answer: D. Each breakdown changes arrivals.

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5. Customers leaving because the line is long are…

A. Jockeying
B. Balking/reneging
C. Batching
D. Blocking

Answer: B. Balk vs renege.

Sources

- P3MD MK4 OSCM — Meeting 7: Waiting Line and Simulation (class slides)
- Jacobs & Chase — OSCM Ch. 10; Schelling (1971); KAI Commuter Manggarai case